

Capacity

Friday, October 11, 2024 11:11 AM

MEASURING CAPACITY

Units of capacity measure the space occupied by substances that can be poured (liquid or granular materials, for example). The most often used customary units of capacity are shown in the table.

Unit	Sample Use	Comparing Units
fluid ounce (fl oz)	 1 fl oz	$1 \text{ fl oz} = \frac{1}{8} \text{ c}$
cup (c)	 1 cup 8 fl oz	$1 \text{ c} = 8 \text{ fl oz}$
pint (pt)	 Milk	$1 \text{ pt} = 2 \text{ c}$
quart (qt)	 motor oil	$1 \text{ qt} = 2 \text{ pt}$
gallon (gal)		$1 \text{ gal} = 4 \text{ qt}$

Converting: Smaller ↔ Larger

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CONVERTING TO LARGER UNITS

Example Jim has 47 fl oz of juice left in a carton. Express this amount in cups and fluid ounces.

Step 1

Set up the problem.

$$47 \text{ fl oz} = \underline{\hspace{1cm}} \text{ c } \underline{\hspace{1cm}} \text{ fl oz}$$

Jim has **5 c 7 fl oz** of juice.

Step 2

To change to a larger unit,
divide by the number of
smaller units.
(8 fl oz = 1 c)

$$47 \text{ fl oz} \div 8 = 5 \text{ R}7$$

Step 3

Write the whole number
as the larger unit and the
remainder as the smaller unit.

$$47 \text{ fl oz} = 5 \text{ c } 7 \text{ fl oz}$$

CONVERTING TO SMALLER UNITS

Example Soo Jung has $1\frac{1}{4}$ qt of applesauce. She needs 38 fl oz. Does she have enough?

Step 1

Set up the problem.

$$1\frac{1}{4} \text{ qt} = \underline{\hspace{1cm}} \text{ fl oz}$$

Step 2

Find out how many fluid
ounces there are in 1 qt.

$$\begin{aligned} 1 \text{ qt} &= 2 \text{ pt} \\ 2 \text{ pt} &= 4 \text{ c} \\ 4 \text{ c} &= 32 \text{ fl oz} \end{aligned}$$

Step 3

To change to a smaller unit,
multiply by the number of
smaller units (1 qt = 32 fl oz).

$$1\frac{1}{4} \times 32 = \frac{5}{4} \times \frac{32}{1} = 40 \text{ fl oz}$$

Yes, Soo Jung has enough applesauce (40 fl oz).

Change units as indicated.

1. $24 \text{ fl oz} = \underline{\hspace{1cm}} \text{ c}$

$5 \text{ c} = \underline{\hspace{1cm}} \text{ pt } \underline{\hspace{1cm}} \text{ c}$

$7 \text{ pt} = \underline{\hspace{1cm}} \text{ qt } \underline{\hspace{1cm}} \text{ pt}$

$14 \text{ qt} = \underline{\hspace{1cm}} \text{ gal } \underline{\hspace{1cm}} \text{ qt}$

2. $2\frac{1}{8} \text{ c} = \underline{\hspace{1cm}} \text{ fl oz}$

$10 \text{ pt} = \underline{\hspace{1cm}} \text{ c}$

$3\frac{3}{4} \text{ gal} = \underline{\hspace{1cm}} \text{ qt}$

$5\frac{1}{2} \text{ qt} = \underline{\hspace{1cm}} \text{ c}$

Adding & Subtracting Capacity

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ADDING OR SUBTRACTING UNITS OF CAPACITY

Example Eileen has a 12-gal-2-qt container. To fill the container, she adds 8 gal 3 qt of water. How much water was in the container before she filled it?

Step 1

Set up the problem.

$$\begin{array}{r} 12 \text{ gal } 2 \text{ qt} \\ - 8 \text{ gal } 3 \text{ qt} \\ \hline \end{array}$$

Step 2

Estimate first by subtracting the larger units.

$$\begin{array}{l} 12 \text{ gal} - 8 \text{ gal} = 4 \text{ gal} \\ \text{The answer should be} \\ \text{slightly less than 4 gal.} \end{array}$$

Step 3

Regroup, if necessary.

$$\begin{array}{l} 12 \text{ gal } 2 \text{ qt} = \\ 11 \text{ gal} + 4 \text{ qt} + 2 \text{ qt} = \\ 11 \text{ gal } 6 \text{ qt} \end{array}$$

Step 4

Add or subtract the units separately. Simplify, if necessary.

$$\begin{array}{r} 11 \text{ gal } 6 \text{ qt} \\ - 8 \text{ gal } 3 \text{ qt} \\ \hline 3 \text{ gal } 3 \text{ qt} \end{array}$$

3 gal 3 qt is slightly less than 4 gal. The container held **3 gal 3 qt** of water.

Add or subtract the following capacities.

3. $\begin{array}{r} 2 \text{ c. } 6 \text{ fl oz} \\ + 3 \text{ c. } 3 \text{ fl oz} \\ \hline \end{array}$

$$\begin{array}{r} 2 \text{ pt } 1 \text{ c} \\ + 2 \text{ pt } 1 \text{ c} \\ \hline \end{array}$$

$$\begin{array}{r} 1 \text{ qt } 1 \text{ pt} \\ + 3 \text{ qt } 1 \text{ pt} \\ \hline \end{array}$$

$$\begin{array}{r} 4 \text{ gal } 3 \text{ qt} \\ + 3 \text{ gal } 2 \text{ qt} \\ \hline \end{array}$$

4. $\begin{array}{r} 6 \text{ c. } 5 \text{ fl oz} \\ - 3 \text{ c. } 2 \text{ fl oz} \\ \hline \end{array}$

$$\begin{array}{r} 8 \text{ c } 4 \text{ fl oz} \\ - 4 \text{ c } 7 \text{ fl oz} \\ \hline \end{array}$$

$$\begin{array}{r} 5 \text{ pt } 0 \text{ c} \\ - 2 \text{ pt } 1 \text{ c} \\ \hline \end{array}$$

$$\begin{array}{r} 10 \text{ gal } 1 \text{ qt} \\ - 3 \text{ gal } 3 \text{ qt} \\ \hline \end{array}$$

Applications

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Solve the following word problems.

5. Yi had a 3-qt.-1-pt bottle of water. He used 2 qt 1 pt. How much water was left?
6. To make instant cider, mix 6–8 fl oz of hot water with a pouch of mix. If Dave adds 3 c of water to 4 pouches of mix, is that enough water?
7. Amanda used 3 c of milk from a 1-gal container. How much milk is left?
8. **Chart** Create a chart listing the number of fluid ounces and cups in a pint, quart, and gallon.